NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

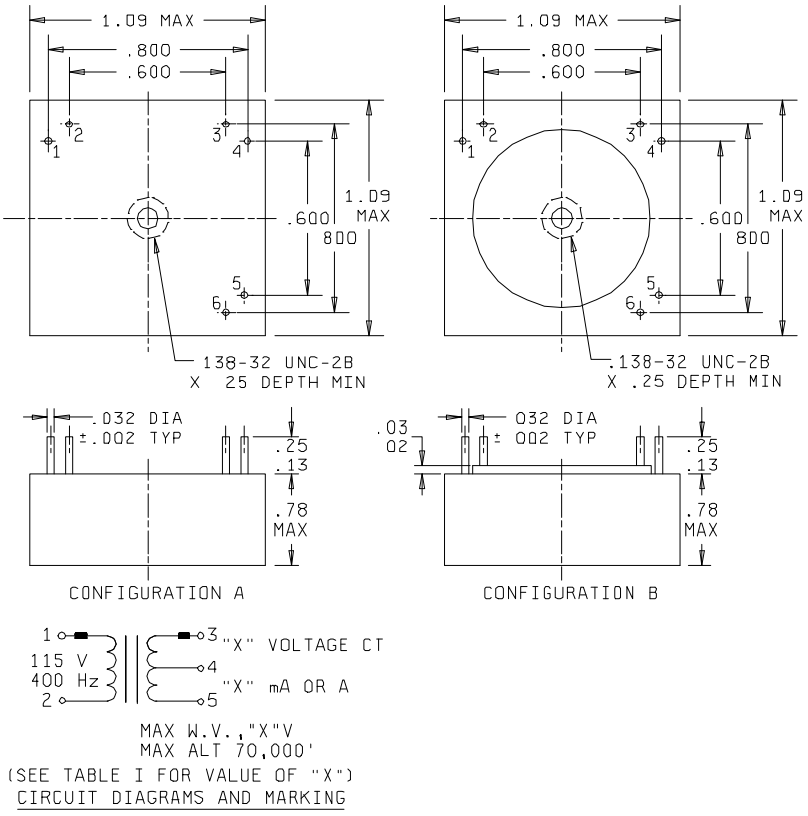
MIL-PRF-27/41B
7 February 1984
SUPERSEDING
MIL-T-27/41A
20 October 1976

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, POWER, 6 VOLT AMPERES, 400 HZ,
TF5S03ZZ576 TO TF5S03ZZ619, INCLUSIVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the transformers described herein shall consist of this specification and the latest issue of MIL-T-27.



INCHES	MM
.002	0.05
.02	0.5
.03	0.8
.032	0.81
.13	3.3
.138	3.51
.25	6.4
.600	15.24
.75	19.1
.78	19.8
.800	20.32
1.09	27.2

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on terminal position dimensions is ±.005 (0.13 mm).
4. Marking shall be on the side of the case.
5. A white dot shall be marked on the side opposite the terminals, over terminal 1, in addition to the terminal marking shown.
6. Secondary electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configuration.

(B) denotes changes

FSC 5950

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REQUIREMENTS: (When numbers in parentheses i.e. (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings. See table I.

Primary voltage (1-2): 115 volts ±10 percent, 400 Hz ±5 percent.

Design and construction:

Dimensions and configuration: See figure 1.

Weight: 1.65 ounces, maximum.

Duty cycle: Continuous.

Case: Plastic encapsulated.

Material. Plastic.

Altitude: 70,000 feet, maximum.

Operating temperature range: -55°C to +130°C maximum.

Terminals: Printed circuit pins - tinned brass.

Terminal strength: Method 211 of MIL-STD-202, test condition A, 2.0 pounds.

Dielectric withstanding voltage:

At sea level: 2.8 times the maximum working voltage specified in table I.

At reduced barometric pressure: 1.25 times the maximum working voltage specified in table I.

Electrical characteristics.

No load With 115 volts, 400 Hz across (1-2): Current in (1-2) shall not exceed 6 mA rms.

Rated load: With 115 volts, 400 Hz across (1-2) and a resistive load with current as specified in table I in (3-5), the voltage across (3-5) shall be as specified in table I.

Efficiency: 81 percent minimum at rated load.

Polarity Additive with terminals 2 and 3 connected.

Regulation $\frac{\text{Voltage (no load)} - \text{Voltage (rated load)}}{\text{Voltage (rated load)}} \times 100$ shall not exceed 12 percent.

Phase shift: 1.0 electrical degree between primary and secondary voltage at rated load, maximum.

Temperature rise: 25°C, maximum.

Vibration: MIL-STD-202, method 204, test condition D.

Marking location See figure 1.

Ⓑ QUALITY ASSURANCE PROVISIONS:

Extent of qualification:

Qualification testing and approval to M27/37-44 A or B, M27/42-44 A or B, and M27/46-44 A or B shall be sufficient to grant qualification approval to MIL-T-27/37 through MIL-T-27/47 inclusive, all parts.

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Qualification testing and approval to M27/41-44 shall be sufficient to grant qualification approval to M27/41-01 through M27/41-44.

Part number example:

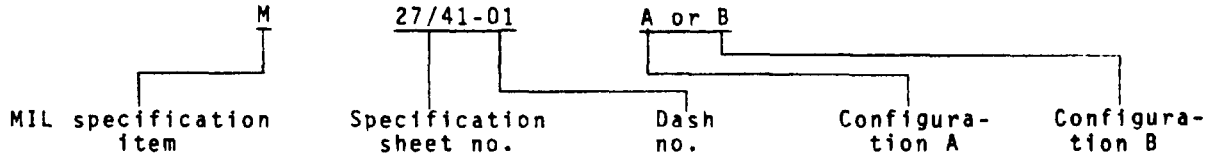


TABLE I. Electrical ratings.

Dash no.	Type designation	Secondary voltage (3-5) (V rms) ±3%	Secondary current (3-5) (amps rms)	Working voltage (V rms max)
01 A or B	576	6	1.100	325
02 "	577	6.3	0.953	"
03 "	578	7	0.857	"
04 "	579	7.5	0.800	"
05 "	580	8	0.750	"
06 "	581	9	0.667	"
07 "	582	10	0.600	"
08 "	583	11	0.545	"
09 "	584	12	0.500	"
10 "	585	12.6	0.476	"
11 "	586	13	0.462	"
12 "	587	15	0.400	"
13 "	588	16	0.375	"
14 "	589	18	0.333	"
15 "	590	20	0.300	"
16 "	591	22	0.272	"
17 "	592	24	0.250	"
18 "	593	26	0.230	"
19 "	594	28	0.214	"
20 "	595	30	0.200	"
21 "	596	33	0.182	"
22 "	597	36	0.167	"
23 "	598	39	0.154	"
24 "	599	43	0.140	"
25 "	600	47	0.128	"
26 "	601	51	0.118	"
27 "	602	56	0.107	"
28 "	603	62	0.097	"
29 "	604	68	0.088	"
30 "	605	75	0.080	"
31 "	606	82	0.073	"
32 "	607	91	0.066	"
33 "	608	100	0.060	"
34 "	609	110	0.055	"
35 "	610	115	0.052	"
36 "	611	120	0.050	"
37 "	612	130	0.046	"
38 "	613	150	0.040	"
39 "	614	180	0.033	"
40 "	615	200	0.030	440
41 "	616	220	0.027	440
42 "	617	240	0.025	440
43 "	618	270	0.022	580
44 "	619	300	0.020	580

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Custodians.
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME
Navy - MC, AS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - ER

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